Manufacturing Execution System Analysis Report

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Executive Summary

Analysis of 13 Color Mismatch defects in Paint and Finish work center on 2026-07-06 identified critical maintenance correlation: Setup/Changeover and Software Updates activities during 11:26-13:34 hours directly preceded defect occurrence. Root cause distribution shows Process Variation (4 defects, 31%) as primary driver, followed by Machine Calibration (3 defects, 23%). Pai-32 exhibits concerning OEE of 0.42 with Availability at 0.72, indicating significant operational challenges. Immediate calibration recovery and maintenance-quality integration are critical to prevent recurrence.

Report Title

Color Mismatch Defect Analysis & Action Plan Report

Analysis Date

2026-07-06

Work Center

Paint and Finish (Pai-31, Pai-32)

Analysis Scope

Oee Analysis: Disabled

Downtime Analysis: Disabled

Changeover Analysis: Disabled

Maintenance Correlation: Enabled

Key Findings

- {'finding': '13 Color Mismatch defects concentrated in Paint work center (Pai-31, Pai-32) on single date 2026-07-06', 'source': 'Maintenance Correlation Analysis', 'certainty': 'HIGH', 'reason': 'Defect data directly provided in analysis scope'}
- {'finding': 'Maintenance events (Setup/Changeover, Software Updates) during 11:26-13:34 hours temporally preceded all defects, indicating direct maintenance-quality correlation', 'source': 'Maintenance Correlation Analysis', 'certainty': 'HIGH', 'reason': 'Maintenance event timing explicitly linked to defect window in analysis results'}
- {'finding': 'Process Variation is dominant root cause (4 defects, 31%), followed by Machine Calibration loss (3 defects, 23%), suggesting post-maintenance calibration drift', 'source': 'Root Cause Analysis within Maintenance Correlation', 'certainty': 'HIGH', 'reason': 'Root cause distribution provided in analysis data'}
- {'finding': 'Pai-32 machine shows critically low OEE of 0.42 with Availability at 0.72 (28% downtime), indicating equipment reliability issues that may amplify calibration sensitivity', 'source': 'Machine Performance Metrics (Maintenance Correlation enabled scope)', 'certainty': 'HIGH', 'reason': 'OEE and Availability metrics explicitly provided for Pai-32'}
- {'finding': 'Material Defect (2 defects) and Design Issue (2 defects) represent secondary contributors requiring supplier audit and design review respectively', 'source': 'Root Cause Distribution Analysis', 'certainty': 'MEDIUM', 'reason': 'Root causes identified but limited context on material batch traceability and design specification details'}

Supporting Data

Table Title: Color Mismatch Defect Root Cause Distribution & Machine Performance

Rows

{'root_cause': 'Process Variation', 'defect_count': 4, 'percentage': '31%', 'primary_machine': 'Pai-31/Pai-32', 'action_priority': 'Immediate - SPC Implementation'}

{'root_cause': 'Machine Calibration', 'defect_count': 3, 'percentage': '23%', 'primary_machine': 'Pai-32', 'action_priority': 'Immediate - Post-Maintenance Calibration'}

{'root_cause': 'Material Defect', 'defect_count': 2, 'percentage': '15%', 'primary_machine': 'Pai-31/Pai-32', 'action_priority': 'Short-term - Supplier Audit'}

{'root_cause': 'Operator Error', 'defect_count': 2, 'percentage': '15%', 'primary_machine': 'Pai-31/Pai-32', 'action_priority': 'Immediate - Retraining'}

{'root_cause': 'Design Issue', 'defect_count': 2, 'percentage': '15%', 'primary_machine': 'Pai-31/Pai-32', 'action_priority': 'Short-term - Design Review'}

{'metric': 'OEE (Pai-32)', 'value': '0.42', 'benchmark': 'Target ≥0.85', 'gap': '-0.43', 'implication': 'Critical equipment performance issue'}

{'metric': 'Availability (Pai-32)', 'value': '0.72', 'benchmark': 'Target ≥0.90', 'gap': '-0.18', 'implication': '28% downtime - investigate root causes'}

{'metric': 'Quality (Pai-32)', 'value': '0.86', 'benchmark': 'Target ≥0.95', 'gap': '-0.09', 'implication': '14% defect rate - calibration improvement needed'}

{'metric': 'Maintenance Event Window', 'value': '11:26-13:34 hours', 'defect_occurrence': 'All 13 defects', 'correlation_strength': '100%', 'implication': 'Direct maintenance-quality correlation confirmed'}

{'metric': 'Defect Concentration Date', 'value': '2026-07-06', 'affected_machines': 'Pai-31, Pai-32', 'total_defects': '13', 'analysis_scope': 'Maintenance Correlation enabled'}

Gaps And Missing Data

- {'gap': 'OEE Analysis disabled - cannot determine Performance and Quality component breakdown for Pai-31; only Pai-32 metrics available', 'impact': 'Limited visibility into whether Pai-31 has similar calibration issues or if Pai-32 is isolated problem', 'recommendation': 'Enable OEE Analysis in future assessments to compare both machines comprehensively'}
- {'gap': 'Downtime Analysis disabled - cannot identify specific downtime causes contributing to Pai-32 Availability loss (0.72)', 'impact': 'Cannot determine if downtime is maintenance-related, equipment failure, or scheduling issue', 'recommendation': 'Enable Downtime Analysis to correlate maintenance events with availability gaps'}
- {'gap': 'Changeover Analysis disabled - cannot assess if Setup/Changeover activities during 11:26-13:34 hours involved product changeovers that may have triggered calibration loss', 'impact': 'Limited understanding of changeover-specific calibration procedures and their effectiveness', 'recommendation': 'Enable Changeover Analysis to evaluate changeover-specific quality impacts'}
- {'gap': 'Material batch traceability not provided - cannot confirm if Material Defect (2 defects) originated from single supplier batch or multiple sources', 'impact': 'Supplier audit scope may be incomplete; cannot isolate material quality issue', 'recommendation': 'Obtain material batch records for 2026-07-06 production to enable targeted supplier corrective action'}

- {'gap': 'Design specification tolerances not provided - cannot assess if Design Issue (2 defects) reflects unachievable specifications or process capability gap', 'impact': 'Design review may recommend unnecessary changes if process capability is actually sufficient', 'recommendation': 'Provide design color matching tolerances and current process capability data (Cpk) for design-process alignment assessment'}

Action Plan Summary

Immediate Actions

{'action': 'Emergency Machine Calibration Recovery', 'timeline': '0-7 days', 'resources': '2 FTE (Maintenance Lead + Paint Technician), calibration equipment', 'success_metric': 'Zero color mismatch defects within 48 hours post-maintenance; calibration drift <0.5%'}

{'action': 'Maintenance Event Documentation & Real-time Alerting', 'timeline': '0-14 days', 'resources': '1.5 FTE (MES Administrator + Maintenance Planner), MES configuration', 'success_metric': '100% maintenance events logged; zero untracked maintenance activities'}

{'action': 'Operator Error Reduction - Retraining Program', 'timeline': '0-30 days', 'resources': '1 FTE (Quality Engineer + Training Coordinator), training materials', 'success_metric': 'Operator-related defects reduced to zero within 30 days'}

Short Term Actions

{'action': 'Statistical Process Control (SPC) Implementation', 'timeline': '1-3 months', 'resources': '2 FTE (Process Engineer + Quality Technician), SPC software, measurement equipment', 'success_metric': 'Process Cpk ≥1.33; defect rate <2% by end of month 3'}

{'action': 'Machine Calibration Preventive Program', 'timeline': '1-3 months', 'resources': '1 FTE dedicated Maintenance Technician, calibration equipment, preventive maintenance software', 'success_metric': 'Pai-32 Quality improve from 0.86 to ≥0.95; calibration drift <0.3%'}

{'action': 'Material Defect Investigation & Supplier Audit', 'timeline': '1-3 months', 'resources': '1.5 FTE (Quality Engineer + Procurement), material testing equipment', 'success_metric': 'Material-related defects eliminated; supplier quality score ≥95%'}

{'action': 'Design Issue Assessment & Specification Review', 'timeline': '1-3 months', 'resources': '1.5 FTE (Design Engineer + Process Engineer), design review meetings', 'success_metric': 'Design-process alignment confirmed; design changes implemented if needed'}

Long Term Actions

{'action': 'Machine Availability & OEE Improvement Program (Pai-32)', 'timeline': '3-12 months', 'resources': '2 FTE (Maintenance Manager + Equipment Engineer), predictive maintenance software, capital budget', 'success_metric': 'Pai-32 Availability improve from 0.72 to ≥0.90; OEE from 0.42 to ≥0.65'}

{'action': 'Automated Color Verification System Implementation', 'timeline': '3-12 months', 'resources': '2 FTE (Automation Engineer + Quality Engineer), inline measurement equipment, MES integration', 'success_metric': '100% color defect detection rate; defect escape rate <0.1%'}

{'action': 'Maintenance-Quality Integration Framework', 'timeline': '3-12 months', 'resources': '1 FTE (Maintenance Manager + Quality Manager), process documentation', 'success_metric': 'Zero unplanned quality impacts from maintenance; 100% maintenance pre-approval'}

Resource Allocation

Immediate Phase: {'fte_required': 3.5, 'capital_investment': '$15,000', 'breakdown': 'Calibration equipment, color reference standards, MES configuration'}

Short Term Phase: {'fte_required': 7.0, 'capital_investment': '$45,000', 'breakdown': 'SPC software, measurement equipment, calibration services, material testing equipment'}

Long Term Phase: {'fte_required': 4.0, 'capital_investment': '$120,000', 'breakdown': 'Predictive maintenance software, inline measurement system, equipment diagnostics, automation integration'}

Total 12 Month Investment: {'total_fte': 14.5, 'total_capital': '$180,000', 'roi_expectation': 'Elimination of 13+ defects/day, improved Pai-32 OEE from 0.42 to ≥0.65, reduced maintenance-induced quality impacts'}

Implementation Roadmap

Phase 1 Immediate: Days 1-30: Emergency calibration recovery, maintenance event tracking, operator retraining

Phase 2 Short Term: Months 1-3: SPC implementation, preventive calibration program, supplier audit, design review

Phase 3 Long Term: Months 3-12: Equipment upgrade evaluation, automated quality system, maintenance-quality integration framework

Success Metrics Overall

Defect Reduction: 13 Color Mismatch defects → 0 within 90 days

Pai32 Oee Improvement: 0.42 → ≥0.65 within 12 months

Pai32 Availability Improvement: 0.72 → ≥0.90 within 12 months

Pai32 Quality Improvement: 0.86 → ≥0.95 within 3 months

Process Capability: Establish baseline → Cpk ≥1.33 within 90 days

Maintenance Quality Impact: Eliminate maintenance-induced defects within 30 days

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